Nathan Anderson

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EDUCATION

Georgia Institute of Technology Expected December 2027

Bachelor of Science in Chemical and Biomolecular Engineering | Cumulative GPA: 3.64/4.00

PROCESS ENGINEERING & RESEARCH EXPERIENCE

Filler Lab, Georgia Institute of Technology Atlanta, GA

Undergraduate Researcher (Device Testing & Process Development) August 2025 – Present

- Build and validate a Python/OpenCV workflow converting microscopy videos into traceable CTQ time-series outputs for microdevice velocity, acceleration, radial drift, and process-response metrics
- Develop repeatable test workflow for thin-film printing process development, including substrate preparation, primer/film application, microscopy characterization, and run-to-run comparison
- Design factorial and response-surface DOE to map CTQ outputs against process inputs; use data analysis to reduce trial-and-error and improve process repeatability for semiconductor manufacturing
- Validate printed microdevice performance using 4-point probe, Keithley SMU Id–Vg/Id–Vd sweeps, and microscopy; correlate electrical outputs with process conditions to reduce yield loss by ~30%

Mayo Clinic Rochester, MN

Engineering Intern (Microfluidic Process Development & Automation) May 2025 – August 2025

- Scaled flow-focusing microfluidic encapsulation via 3D-printed tooling; increased throughput by 20x while documenting process parameters, test conditions, and transfer-relevant changes
- Optimized flow ratios and integrated crosslink automation; increased encapsulation yield by ~50% through systematic testing, troubleshooting, and repeatability-focused process adjustments
- Prototyped ESP32/Arduino-controlled thermocycling automation for nanoliter qPCR workflow integrating cell capture, 3-D cell culture, lysis, transcription, and amplification steps
- Built image-analysis-supported test method to quantify capsule size, coalescence, and run-to-run variability; used results to guide process adjustments

OPERATIONS & QUALITY

McDonald's Rochester, MN

Manager May 2021 – August 2022

- Increased cars served/hr 20% and reduced order-to-handoff time 15% by line balancing from POS during peaks
- Launched preventive-maintenance calendar and quality-control list for grills/fryers, reducing downtime ~30%

LEADERSHIP & PROJECTS

Founder and President, *Nanotechnology Club* September 2025 – Present

- Establish multidisciplinary nanotechnology club focused on applications of nanotechnology and nanosystems
- Direct a semester project where teams design and prototype nanosystems in a mini-DOE

Pro-bono Consultant, *Consult Your Community* August 2024 – Present

- Deliver pro-bono consulting for a local cancer-support nonprofit; increase social media engagement +400%
- Translate ambiguous goals into CTQs/KPIs and a scoped workplan with clear success metrics

SKILLS

Process Development: DOE, ANOVA, SPC, CTQ Definition, Control Plans, Process-window Troubleshooting & RCA, FMEA
Bioprocessing Methods: Flow-focusing Microfluidics, Encapsulation, Cell Capture, Lysis, Transcription, qPCR, PCR
Metrology & Characterization: SEM, AFM, XRD, XPS, Raman, Surface Roughness (Sa/Sq), Keithley SMU, Microscopy
Software: JMP, Python (Data Analysis/Automation), MATLAB, Microsoft Suite, Tableau, Arduino, AutoCAD, Aspen Plus